SCIENTIFIC EVALUATION REPORT

BabyVision-v2 on Public Gear Problems

Three Models, Three Replicates, One Frozen Image-Only Route

63.6%	297	33	$226.54
best strict accuracy	attempted trials	timeouts	list-price estimate

Report date	August 16, 2026
Dataset	33 public base questions; six PNG views per task
Models	`qwen3.7-plus`, `claude-opus-5`, `claude-fable-5`
Condition	BabyVision-v2 `cc2d-6`; no MCP; one attempt; zero retries
Replication	Three runs per model; 99 attempts per model
Primary endpoint	Strict full-credit accuracy over all planned attempts
Statistics	Wilson interval; 20,000-resample question-cluster bootstrap; McNemar sensitivity
Evidence audit	Hidden paths excluded; no raw CAD/scene assets available
Status	Complete, offline regraded, response-model and credential audited

The public snapshot cannot support a matched MCP comparison. Reconstructed geometry was not substituted for absent provenance-matched source CAD.

Abstract

Background. Multi-view gear questions combine visual correspondence, repeated-part counting, categorical recognition, and transmission arithmetic. **Methods.** BabyVision-v2 evaluated three exact model routes on the same 33 public base tasks with six views per task and three one-attempt replicates per model. Timeouts remained incorrect in the primary denominator. Comparative confidence intervals used a question-cluster bootstrap. **Results.** The highest strict accuracy was 63/99 (63.6%; question-cluster 95% CI 47.5-78.8%), observed for Claude Opus 5. Across all models, the study recorded 114,762,672 input tokens, 3,325,592 output tokens, 33 timeouts, and a reconstructed list-price total of $226.54. **Conclusion.** The controlled matrix separates answer quality, repeatability, and execution reliability. MCP effectiveness remains non-estimable because the frozen packages contain no matched raw CAD or scene assets.

Interpretation. Every planned attempt remains in the strict denominator. Timeout-adjusted accuracy is a sensitivity view. Costs apply published rates to native token fields and are not invoices.

Contents

1 Introduction

Mechanical assembly questions are systems tasks rather than single-image recognition problems. An agent must preserve object identity across camera angles, distinguish repeated parts, count teeth or components, recognize gear shape, and sometimes infer a multi-stage transmission ratio. A trustworthy evaluation must therefore control which evidence is visible, keep scoring authority separate, and retain operational failures instead of hiding them inside a successful-answer denominator.

BabyVision-v2 provides this control through manifest-driven staging, isolated Harbor execution, declared final-answer contracts, deterministic verifier-side scoring, and preserved trajectories. The earlier Fable study contributed two no-MCP repeatability runs on the public gear subset. The present study extends that design to three exact model routes and three replicate indices per model while preserving the same frozen evidence and strict operational policy.

The objectives were to compare strict and timeout-adjusted accuracy, quantify question-cluster uncertainty and run-to-run repeatability, identify question-type and answer-level failure modes, audit response-model identity and native usage, and establish a reproducible publication bundle. A matched MCP comparison was planned conceptually but not executed because the public source packages contain rendered views without provenance-matched STEP, assembly, mesh, or scene assets.

2 Methods

2.1 Frozen scope and evidence boundary

The dataset contained 33 public base questions: one axis-angle question, eight component-count questions, one gear-type question, thirteen tooth-count questions, and ten transmission-ratio questions. Each task staged six source-defined PNG views, producing 198 unique images. The same manifest and images were used by all model replicates. Hidden packages, dot-prefixed paths, raw-asset-like files, and private answer keys were excluded from agent environments. Ground truth entered only through the scorer-side task metadata.

2.2 Execution and replication

The exact model identifiers were `qwen3.7-plus`, `claude-opus-5`, and `claude-fable-5`. Live probes confirmed those identifiers before inference. Each model contributed three replicate indices and 99 planned attempts. The two completed Fable replicates were reused verbatim; Fable C and all Qwen and Opus replicates were newly executed. Each task had one attempt, a 600-second deadline, zero automatic retries, no fallback model, and the same BabyVision-v2 `cc2d-6` Docker environment. Qwen used Alibaba Cloud Model Studio's Anthropic-compatible endpoint; Opus and Fable used the Anthropic Messages API.

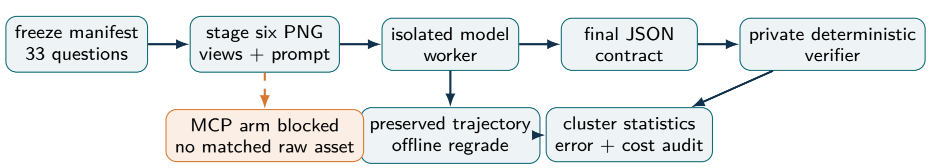

Figure 1: Evaluation and information-flow schematic. The model sees only the public question and six images. Scorer-side ground truth is private, and the MCP branch remains gated on matched raw geometry.

2.3 Endpoints and statistical analysis

The primary endpoint was strict full-credit accuracy, defined as answer_score = 1.0 over every planned attempt. Agent, environment, verifier, or cancellation timeouts remained incorrect. A labeled sensitivity analysis removed timeout records from the denominator. Wilson intervals describe the attempt-level proportions. Because each question appears three times per model, comparative uncertainty used 20,000 bootstrap samples clustered by question with deterministic seed 20260816. Pairwise aligned McNemar tests were retained as sensitivity analyses. Repeatability used the three within-model decisions per question to calculate pairwise correctness agreement, unanimous outcomes, mixed outcomes, and Fleiss' kappa.

Native session records were deduplicated by message ID and summed across prompt, cache-creation, cache-read, and output fields. Qwen's global below-256K list rates were \$0.276 per million standard input tokens and \$1.101 per million output tokens; explicit five-minute cache creation was valued at 125% and reads at 10% of the input rate. The largest observed Qwen request contained 64,176 input tokens, below the tier boundary. Opus 5 used \$5/\$25 per million input/output tokens, and Fable 5 used \$10/\$50. Anthropic five-minute and one-hour cache writes used 125% and 200%, and reads used 10% of the model's input rate. The calculation is descriptive and is not an invoice.

3 Results

3.1 Trial accounting and overall accuracy

All three models contributed 99 planned attempts, yielding 297 trials. Table 1 reports strict accuracy, question-cluster uncertainty, timeout burden, and the non-timeout sensitivity view. Claude Opus 5 produced the highest strict point estimate.

Table 1: Overall outcomes by model.

Model	Correct	Incorrect	Strict acc.	Cluster 95% CI	Timeouts	Non-timeout acc.
Qwen 3.7 Plus	22	77	22.2%	10.1-35.4%	4	23.2%
Claude Opus 5	63	36	63.6%	47.5-78.8%	14	74.1%
Claude Fable 5	59	40	59.6%	43.4-74.7%	15	70.2%

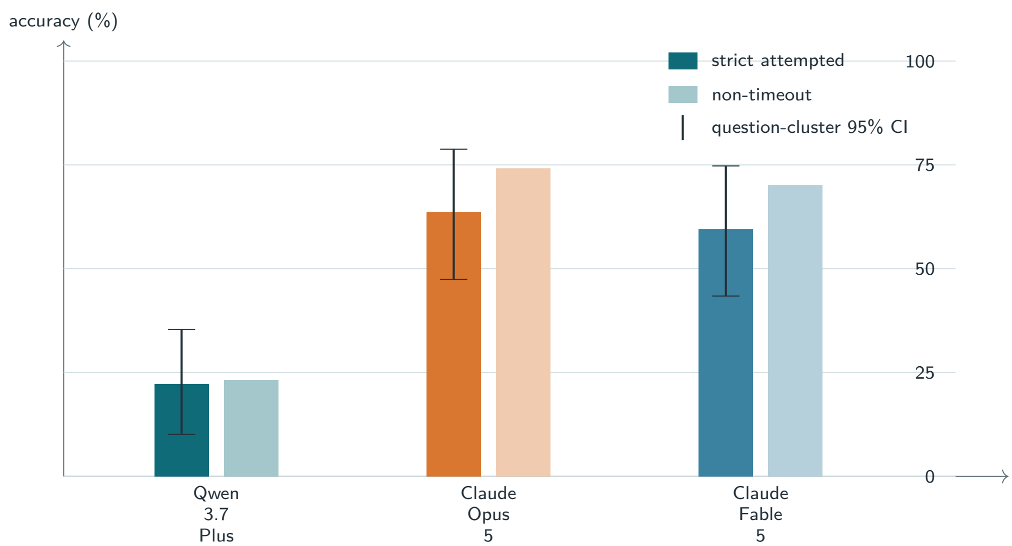

Figure 2: Strict attempted accuracy and timeout-adjusted sensitivity. Whiskers on the strict bars show the 95% question-cluster bootstrap interval.

3.2 Question-type performance

Performance varied across the five mechanical question types (Table 2). Axis-angle and gear-type cells each summarize only three attempts on one unique question, whereas tooth-count and transmission-ratio cells summarize 39 and 30 attempts per model, respectively.

Table 2: Strict accuracy by question type.

Type	Attempts/model	Qwen 3.7 Plus	Claude Opus 5	Claude Fable 5
Axis angle	3	100.0%	100.0%	100.0%
Component count	24	45.8%	75.0%	75.0%
Gear type	3	0.0%	0.0%	0.0%
Tooth count	39	2.6%	66.7%	59.0%
Transmission ratio	30	23.3%	53.3%	50.0%

3.3 Repeatability and pairwise differences

Within-model repeatability is shown in Table 3. Pairwise agreement treats every pair of replicate decisions on the same question as one comparison. Fleiss' kappa adjusts that agreement for the marginal correct/incorrect distribution.

Table 3: Within-model repeatability over 33 questions.

Model	Pairwise agreement	Fleiss κ	Unanimous correct	Unanimous wrong	Mixed
Qwen 3.7 Plus	89.9%	0.708	5	23	5
Claude Opus 5	91.9%	0.825	19	10	4
Claude Fable 5	87.9%	0.748	17	10	6

Table 4 presents paired model differences. The cluster-bootstrap interval is the principal uncertainty result; McNemar's exact test exposes the aligned discordant counts but does not remove repeated-question dependence.

Table 4: Pairwise model comparisons.

Comparison	Difference (pp)	Cluster 95% CI	Left only	Right only	McNemar p
Qwen 3.7 Plus → Claude Opus 5	41.4	24.2-58.6%	3	44	0.0000
Qwen 3.7 Plus → Claude Fable 5	37.4	21.2-53.5%	3	40	0.0000
Claude Opus 5 → Claude Fable 5	-4.0	-11.1-1.0%	8	4	0.3877

3.4 Errors, usage, latency, and cost

The deterministic audit separated timeouts and other operational failures from missing JSON, numeric direction errors, reciprocal-ratio errors, categorical mismatches, and other substantive wrong answers. Table 5 gives the model-level counts; the supplementary CSV retains every submitted and expected answer.

Table 5: Incorrect-attempt taxonomy.

Error class	Qwen 3.7 Plus	Claude Opus 5	Claude Fable 5
categorical mismatch	3	3	3
numeric overestimate	9	7	9
numeric underestimate	60	12	13
operational other	1	0	0
operational timeout	4	14	15

3.5 Targeted question audit

Table 6 shows the ten lowest-scoring questions across the nine aligned attempts. The counts of questions missed in all three within-model runs were Qwen 3.7 Plus: 23; Claude Opus 5: 10; Claude Fable 5: 10. Questions missed in all nine attempts: Model02_QA03, Model03_QA03, Model04_QA04, Model06_QA01, Model07_QA03, Model08_QA01, Model08_QA02, Model08_QA03, Model08_QA04.

Table 6: Lowest-scoring questions across all models and replicates.

Task	Type	Correct	Models wrong in all three runs
Model02_QA03	Component count	0/9	Qwen 3.7 Plus, Claude Opus 5, Claude Fable 5
Model03_QA03	Tooth count	0/9	Qwen 3.7 Plus, Claude Opus 5, Claude Fable 5
Model04_QA04	Tooth count	0/9	Qwen 3.7 Plus, Claude Opus 5, Claude Fable 5
Model06_QA01	Component count	0/9	Qwen 3.7 Plus, Claude Opus 5, Claude Fable 5
Model07_QA03	Transmission ratio	0/9	Qwen 3.7 Plus, Claude Opus 5, Claude Fable 5
Model08_QA01	Tooth count	0/9	Qwen 3.7 Plus, Claude Opus 5, Claude Fable 5
Model08_QA02	Tooth count	0/9	Qwen 3.7 Plus, Claude Opus 5, Claude Fable 5
Model08_QA03	Transmission ratio	0/9	Qwen 3.7 Plus, Claude Opus 5, Claude Fable 5
Model08_QA04	Gear type	0/9	Qwen 3.7 Plus, Claude Opus 5, Claude Fable 5
Model02_QA01	Transmission ratio	1/9	Qwen 3.7 Plus, Claude Opus 5

Native telemetry and reconstructed list-price costs appear in Table 7. Every response-model record was compared with the requested identifier; deviations, if any, are recorded in the machine-readable summary and trial audit.

Table 7: Native usage, latency, and reconstructed cost.

Model	Input tokens	Output tokens	Calls	Median sec	P90 sec	Cost USD
Qwen 3.7 Plus	9,092,202	387,762	338	61.7	191.6	1.1991
Claude Opus 5	59,524,843	1,607,258	1,579	173.0	622.8	85.4007
Claude Fable 5	46,145,627	1,330,572	1,342	229.7	623.1	139.9442

The reconstructed full-matrix cost was \$226.54. The 231 newly executed trials accounted for \$136.0556, while the 66 reused Fable trials accounted for \$90.4884. The new expansion remained within the predeclared \$200 soft ceiling.

4 Discussion

The study separates answer quality from execution reliability and replicate stability. Claude Opus 5 had the highest strict point estimate, but interpretation depends on the question-cluster interval and paired discordance because the 99 model attempts arise from only 33 unique questions. Timeout-adjusted percentages indicate how execution failures affect the denominator without erasing substantive visual or arithmetic mistakes.

Question-type results identify the mechanical operations responsible for most errors. Counting failures implicate cross-view correspondence and occlusion; transmission-ratio failures can combine tooth counting, gear-contact interpretation, direction conventions, and arithmetic; categorical failures can reflect insufficient view integration. The complete audit makes these mechanisms inspectable at task level rather than inferring them from aggregate accuracy alone.

Repeatability is a deployment property as well as a statistical concern. A model with a strong point estimate but many mixed three-run question profiles is less predictable than one with similar accuracy and higher unanimity. Conversely, unanimous errors identify systematic blind spots that are unlikely to be repaired by merely repeating the same prompt.

Cost and latency are descriptive engineering outcomes. List-price reconstruction supports accuracy-per-dollar comparisons, but actual billing can differ with promotions, cache policy, contracts, taxes, or service routing. The retained native usage and exact response-model counts provide the evidence needed to recompute costs under other rate assumptions.

5 Limitations and reproducibility

The public base set is small and uneven across question types. Replicate attempts are clustered observations, not 99 independent questions. Model services, identifiers, and prices can change after the recorded probe date. The experiment also does not estimate an MCP effect. No provenance-matched raw CAD, mesh, assembly, or scene asset exists for the 33 frozen questions, and reconstructing one from rendered views would introduce a new evidence-generation process.

The expanded plan, task and scoring manifest hashes, BabyVision commit and working-diff fingerprint, Docker image digests, Python and Harbor versions, model probes, run state, preserved trajectories, offline regrades, complete trial JSONL, incorrect audit, statistical tables, and credential scan are retained with this report. The two reused Fable replicates remain byte-identifiable source artifacts, and every new worker has an explicit completion status.

6 Conclusion

BabyVision-v2 produced a controlled three-model comparison on identical six-view public gear evidence. The highest strict accuracy was 63.6%, observed for Claude Opus 5; the cluster intervals, repeatability statistics, error taxonomy, timeout sensitivity, native usage, and cost define the uncertainty and practical meaning of that result. The next causal evidence-access experiment should add provenance-matched raw CAD or scene assets for every task while preserving all other controls.

References

1. UniPat-AI. BabyVision-v2 evaluation pipeline. `https://github.com/UniPat-AI/BabyVision-v2`.
2. AxioForge AI. Gear assemblies QA repository. `https://github.com/axioforgeai/gear-assemblies-qa`.
3. Alibaba Cloud Model Studio. Model pricing. `https://www.alibabacloud.com/help/en/model-studio/model-pricing`.
4. Alibaba Cloud Model Studio. Context cache billing. `https://www.alibabacloud.com/help/en/model-studio/context-cache`.
5. Anthropic. Claude pricing documentation. `https://platform.claude.com/docs/en/about-claude/pricing`.

A Artifact identifiers

Expanded plan	`27071d53d11cc9846f9c6c5fe4330aac5a4ffa3969709ee0c49db0dea6d2d35d`
Task manifest	`297b637d5393df3ebce3d33cb9eeaea3aef17d05566ea23625b31383c996e41e`
Private scoring manifest	`2d5814139b9866549862c77f11a962a2df2749fc79f959f588e13f6cb9e1066d`

The complete incorrect-answer audit and machine-readable summary are distributed beside this manuscript.